Knowledge organiser · a scientific answer

I can make a claim that answers the question.

I can add relevant evidence and a scientific reason.

I can keep my first answer and make one pupil-owned improvement.

I can give a scientific answer with evidence and reasoning, then improve it after audience response.

A claim answers the question. Evidence is a useful result or model clue. A reason links that clue to a science idea.

Use claim, evidence and reason together. Keep the first answer. An adult says back what they heard or saw. Then choose a change, or say why you will keep the answer.

The practice rover distances are smooth 46, 44, 45 cm and fabric 18, 20, 19 cm. All three fabric runs are shorter. Friction is the model's science explanation. These practice data are not a new test.

A fair limit says what a result or model cannot show. More words alone do not make a stronger answer. Keep your own meaning when you improve it.

Made from the web lesson SCI-GROW-A2-W05-P2
Version _____

Made by Matt · © Matt Roper · CC BY-NC-SA 4.0: free for classroom use, share and adapt with credit, no resale

Word page

I can make a claim that answers the question.

I can add relevant evidence and a scientific reason.

I can keep my first answer and make one pupil-owned improvement.

Word	Plain bridge	Meaning
claim	my answer	a statement that answers the scientific question
reasoning	why the evidence matters	the scientific idea that connects evidence to a claim
revise	make meaning clearer	change an answer after checking it while keeping its ownership

Claim is my answer. Evidence is the clue I use. Reason is how science links the clue and answer. Revise means check and make my meaning clear.

Made from the web lesson SCI-GROW-A2-W05-P2
Version _____

Made by Matt · © Matt Roper · CC BY-NC-SA 4.0: free for classroom use, share and adapt with credit, no resale

Arrival · Supported

I can make a claim that answers the question.

I can add relevant evidence and a scientific reason.

I can keep my first answer and make one pupil-owned improvement.

Use your book. Point, say, sign, draw, type, write or use an exact scribe. You may pass. Keep your first answer.

Recap: link question, evidence and reason. A claim must stay within what was measured. A model can help while leaving things out.

1. Point to the claim in a three-part answer.
2. Point to the rover numbers used as evidence.
3. Choose the reason that names friction.
4. Complete: My answer became clearer when I added ___.

Made from the web lesson SCI-GROW-A2-W05-P2
Version _____

Made by Matt · © Matt Roper · CC BY-NC-SA 4.0: free for classroom use, share and adapt with credit, no resale

Arrival · Independent

I can make a claim that answers the question.

I can add relevant evidence and a scientific reason.

I can keep my first answer and make one pupil-owned improvement.

Use your book. Point, say, sign, draw, type, write or use an exact scribe. You may pass. Keep your first answer.

Recap: link question, evidence and reason. A claim must stay within what was measured. A model can help while leaving things out.

1. Name the three parts of a strong short answer.
2. Explain why the first answer must be kept.
3. State one useful audience question.
4. Improve: 'Fabric slowed it' by adding evidence.

Made from the web lesson SCI-GROW-A2-W05-P2
Version _____

Made by Matt · © Matt Roper · CC BY-NC-SA 4.0: free for classroom use, share and adapt with credit, no resale

Arrival · Aspire

I can make a claim that answers the question.

I can add relevant evidence and a scientific reason.

I can keep my first answer and make one pupil-owned improvement.

Use your book. Point, say, sign, draw, type, write or use an exact scribe. You may pass. Keep your first answer.

Recap: link question, evidence and reason. A claim must stay within what was measured. A model can help while leaving things out.

1. Distinguish clarification from an adult rewriting the answer.
2. Explain why more words do not always mean stronger evidence.
3. Name one fair limitation of the rover data.
4. State when a pupil might justify no change after audience response.

Made from the web lesson SCI-GROW-A2-W05-P2
Version _____

Made by Matt · © Matt Roper · CC BY-NC-SA 4.0: free for classroom use, share and adapt with credit, no resale

Starter · first idea

I can make a claim that answers the question.

I can add relevant evidence and a scientific reason.

I can keep my first answer and make one pupil-owned improvement.

Which addition would most improve: 'The rover went farther on smooth surface'?

A. Add the repeated distances and explain friction

B. Add three unrelated space facts

C. Use bigger lettering

Choose one idea before the model. Point, say or sign a reason. Keep your first idea so you can check it later.

Made by Matt · © Matt Roper · CC BY-NC-SA 4.0: free for classroom use, share and adapt with credit, no resale

CLAIM · answer the question

I can make a claim that answers the question.

I can add relevant evidence and a scientific reason.

I can keep my first answer and make one pupil-owned improvement.

CLAIM

The rover travels farther on the smooth surface.

Keep this first answer. It needs the next evidence and reason to support it.

Made from the web lesson SCI-GROW-A2-W05-P2
Version _____

Made by Matt · © Matt Roper · CC BY-NC-SA 4.0: free for classroom use, share and adapt with credit, no resale

EVIDENCE · use a relevant result

I can make a claim that answers the question.

I can add relevant evidence and a scientific reason.

I can keep my first answer and make one pupil-owned improvement.

EVIDENCE

smooth 46, 44, 45 cm · fabric 18, 20, 19 cm

every fabric result is lower

These are worked practice data with units and repeats.

Compare all three runs on each surface. Each fabric run is shorter than each smooth run. That is a repeated pattern in these practice data. It is not a new test.

Made from the web lesson SCI-GROW-A2-W05-P2
Version _____

Made by Matt · © Matt Roper · CC BY-NC-SA 4.0: free for classroom use, share and adapt with credit, no resale

REASON · explain the science

I can make a claim that answers the question.

I can add relevant evidence and a scientific reason.

I can keep my first answer and make one pupil-owned improvement.

REASON

fabric creates more friction, opposing motion,

so the rover slows and stops sooner

Use the science idea to say why the pattern fits the claim. Keep the model limit in view.

Made from the web lesson SCI-GROW-A2-W05-P2
Version _____

Made by Matt · © Matt Roper · CC BY-NC-SA 4.0: free for classroom use, share and adapt with credit, no resale

I do · worked answer

I can make a claim that answers the question.

I can add relevant evidence and a scientific reason.

I can keep my first answer and make one pupil-owned improvement.

Use your book. Point, say, sign, draw, type, write or use an exact scribe. You may pass. Keep your first answer.

Worked practice answer: 'The rover went farther on the smooth surface.' Keep it visible as the pupil's starting Voice; these practice data are not next week's test.

Add worked evidence: smooth 46, 44, 45 cm; fabric 18, 20, 19 cm. Select only the values that answer the question.

Add reasoning: the fabric produced more friction, opposing motion, so the rover travelled less far.

Keep the first answer so we can see what you chose. Hear the adult say back your science. Then choose a change, or say why you will keep the answer.

One answer shows what you can do today. It cannot show all that you know. A point, sign, drawing or exact scribe can show the same science.

Made by Matt · © Matt Roper · CC BY-NC-SA 4.0: free for classroom use, share and adapt with credit, no resale

Claim, evidence and reason builder

I can make a claim that answers the question.

I can add relevant evidence and a scientific reason.

I can keep my first answer and make one pupil-owned improvement.

Put each sentence in Claim, Evidence, Reason or Weak. Two sentences are weak. Say why.

1. The rover travels farther on the smooth surface.
2. Smooth 46, 44, 45 cm; fabric 18, 20, 19 cm.
3. Fabric creates more friction, which opposes motion, so the rover stops sooner.
4. The rover went farther because smooth is better.
5. It went 46 cm once on smooth.

An adult asks: How does friction explain the difference? Keep your answer and choose your own next move.

Made from the web lesson SCI-GROW-A2-W05-P2
Version _____

Made by Matt · © Matt Roper · CC BY-NC-SA 4.0: free for classroom use, share and adapt with credit, no resale

We do · choose and explain

I can make a claim that answers the question.

I can add relevant evidence and a scientific reason.

I can keep my first answer and make one pupil-owned improvement.

Which revision is scientifically strongest?

A. The rover went farther on smooth surface: 44–46 cm versus 18–20 cm, because fabric produced more friction.

B. The rover was excellent, amazing, fast and very interesting on the table.

C. The Earth rotates once in about 24 hours, and the Moon reflects sunlight.

Show one choice and the clue you used. A retry helps us see the next step. The adult names back the science heard without adding to your words.

Made from the web lesson SCI-GROW-A2-W05-P2
Version _____

Made by Matt · © Matt Roper · CC BY-NC-SA 4.0: free for classroom use, share and adapt with credit, no resale

I do 2 · an audience and a next answer

I can make a claim that answers the question.

I can add relevant evidence and a scientific reason.

I can keep my first answer and make one pupil-owned improvement.

The adult says back what they heard: 'I heard your claim and the smooth-surface evidence.'

The adult asks one check: 'How does friction explain the difference?' You choose the next move.

Keep your first answer. Add your own next answer beside it. Say what is clearer now.

Keep the first answer so we can see what you chose. Hear the adult say back your science. Then choose a change, or say why you will keep the answer.

One answer shows what you can do today. It cannot show all that you know. A point, sign, drawing or exact scribe can show the same science.

Made by Matt · © Matt Roper · CC BY-NC-SA 4.0: free for classroom use, share and adapt with credit, no resale

Answer lab · forces

I can make a claim that answers the question.

I can add relevant evidence and a scientific reason.

I can keep my first answer and make one pupil-owned improvement.

Match the answer and its limit to Forces, Space or Climate. Explain how the clue supports the answer.

FORCES ANSWER · Why less far on fabric?

Practice evidence: smooth 46, 44, 45 cm; fabric 18, 20, 19 cm. Greater fabric friction opposed motion more.

FORCES LIMIT · one small rover

One rover and two classroom surfaces cannot represent every vehicle or terrain.

Keep one answer with its limit. Hear what the adult received. Then choose what to add, make clear or keep.

Made from the web lesson SCI-GROW-A2-W05-P2
Version _____

Made by Matt · © Matt Roper · CC BY-NC-SA 4.0: free for classroom use, share and adapt with credit, no resale

Answer lab · space

I can make a claim that answers the question.

I can add relevant evidence and a scientific reason.

I can keep my first answer and make one pupil-owned improvement.

Match the answer and its limit to Forces, Space or Climate. Explain how the clue supports the answer.

SPACE ANSWER · Why day and night?

Model evidence: a marked place turns into and out of the lamp's light. Earth rotates, so day and night repeat.

SPACE LIMIT · simplified lamp and globe

The model is not to scale and fixing the lamp is an assumption; it shows how rotation explains the pattern.

Keep one answer with its limit. Hear what the adult received. Then choose what to add, make clear or keep.

Made from the web lesson SCI-GROW-A2-W05-P2
Version _____

Made by Matt · © Matt Roper · CC BY-NC-SA 4.0: free for classroom use, share and adapt with credit, no resale

Answer lab · climate

I can make a claim that answers the question.

I can add relevant evidence and a scientific reason.

I can keep my first answer and make one pupil-owned improvement.

Match the answer and its limit to Forces, Space or Climate. Explain how the clue supports the answer.

CLIMATE ANSWER · Why can extra gas warm Earth?

Energy route: Earth gives out infrared energy. Extra greenhouse gas absorbs more of it. Some energy comes back towards the surface. Earth loses energy more slowly, so the global average rises.

CLIMATE LIMIT · arrows simplify a complex system

The arrows show which way energy goes. They do not show real amounts, times or all climate feedbacks.

Keep one answer with its limit. Hear what the adult received. Then choose what to add, make clear or keep.

Made by Matt · © Matt Roper · CC BY-NC-SA 4.0: free for classroom use, share and adapt with credit, no resale

Science glitch · repair the claim

I can make a claim that answers the question.

I can add relevant evidence and a scientific reason.

I can keep my first answer and make one pupil-owned improvement.

A longer answer is always a stronger scientific answer.

A. Keep it: the number of words decides quality.

B. Repair it: strength comes from a relevant claim, evidence and reason.

C. Change it: every answer must be one sentence only.

This claim is a challenge to check. It is not the teaching rule. Say which choice keeps a useful science answer.

Made from the web lesson SCI-GROW-A2-W05-P2
Version _____

Made by Matt · © Matt Roper · CC BY-NC-SA 4.0: free for classroom use, share and adapt with credit, no resale

Independent · Supported

I can make a claim that answers the question.

I can add relevant evidence and a scientific reason.

I can keep my first answer and make one pupil-owned improvement.

Use your book. Point, say, sign, draw, type, write or use an exact scribe. You may pass. Keep your first answer.

Choose one embedded question—rover friction, day and night, or extra greenhouse gas and warming—give a first answer, have it received by an adult, then revise or justify keeping it.

Choose one topic answer. Find its claim, evidence and reason. Share it in a way that works for you.

Adult reads back exact choices; pupil selects Add evidence, Clarify reason or Keep and explain.

No forced public speaking or recording. Use an audience of one, pointing, quiet handover or exact adult scribing.

Choose one route within the full 15 minutes. Use paper, cards or an offline document. Nothing is uploaded.

Made by Matt · © Matt Roper · CC BY-NC-SA 4.0: free for classroom use, share and adapt with credit, no resale

Independent · Independent

I can make a claim that answers the question.

I can add relevant evidence and a scientific reason.

I can keep my first answer and make one pupil-owned improvement.

Use your book. Point, say, sign, draw, type, write or use an exact scribe. You may pass. Keep your first answer.

Choose one embedded question—rover friction, day and night, or extra greenhouse gas and warming—give a first answer, have it received by an adult, then revise or justify keeping it.

Answer one of the three embedded questions independently, receive one audience check and revise one part.

Use Claim, Evidence and Reason headings only if needed; fade them after the first success.

No forced public speaking or recording. Use an audience of one, pointing, quiet handover or exact adult scribing.

Choose one route within the full 15 minutes. Use paper, cards or an offline document. Nothing is uploaded.

Made by Matt · © Matt Roper · CC BY-NC-SA 4.0: free for classroom use, share and adapt with credit, no resale

Independent · Aspire

I can make a claim that answers the question.

I can add relevant evidence and a scientific reason.

I can keep my first answer and make one pupil-owned improvement.

Use your book. Point, say, sign, draw, type, write or use an exact scribe. You may pass. Keep your first answer.

Choose one embedded question—rover friction, day and night, or extra greenhouse gas and warming—give a first answer, have it received by an adult, then revise or justify keeping it.

Answer a second topic question. Which evidence is stronger, and why? Explain why you changed your answer or kept it.

Add one limitation without weakening a secure conclusion.

No forced public speaking or recording. Use an audience of one, pointing, quiet handover or exact adult scribing.

Choose one route within the full 15 minutes. Use paper, cards or an offline document. Nothing is uploaded.

Made by Matt · © Matt Roper · CC BY-NC-SA 4.0: free for classroom use, share and adapt with credit, no resale

First answer and next answer

I can make a claim that answers the question.

I can add relevant evidence and a scientific reason.

I can keep my first answer and make one pupil-owned improvement.

Choose rover friction, day and night, or extra greenhouse gas and warming. Keep each part of your own response.

First answer	Next answer or justified no change
_____	_____
_____	_____

My evidence or model clue: _____

What the adult heard or saw: _____

My next move and why: _____

Keep the first answer beside the next. The adult may write your exact words. You own the choice to change or keep them.

Made from the web lesson SCI-GROW-A2-W05-P2
Version _____

Made by Matt · © Matt Roper · CC BY-NC-SA 4.0: free for classroom use, share and adapt with credit, no resale

Exit · Supported

I can make a claim that answers the question.

I can add relevant evidence and a scientific reason.

I can keep my first answer and make one pupil-owned improvement.

Point to what changed between your first and next answer.

Show or say your own response. The adult names back the science they heard or saw. Then choose a next move.

Secure / mixed / misconception / access / method or evidence.

What I said, and what it changed: _____

Next I will: _____

Use your book. Point, say, sign, draw, type, write or use an exact scribe. You may pass. Keep your first answer.

Made from the web lesson SCI-GROW-A2-W05-P2
Version _____

Made by Matt · © Matt Roper · CC BY-NC-SA 4.0: free for classroom use, share and adapt with credit, no resale

Exit · Independent

I can make a claim that answers the question.

I can add relevant evidence and a scientific reason.

I can keep my first answer and make one pupil-owned improvement.

Explain how audience response made your Science clearer.

Show or say your own response. The adult names back the science they heard or saw. Then choose a next move.

Secure / mixed / misconception / access / method or evidence.

What I said, and what it changed: _____

Next I will: _____

Use your book. Point, say, sign, draw, type, write or use an exact scribe. You may pass. Keep your first answer.

Made from the web lesson SCI-GROW-A2-W05-P2
Version _____

Made by Matt · © Matt Roper · CC BY-NC-SA 4.0: free for classroom use, share and adapt with credit, no resale

Exit · Aspire

I can make a claim that answers the question.

I can add relevant evidence and a scientific reason.

I can keep my first answer and make one pupil-owned improvement.

Justify a change or no-change decision using relevance, evidence and limitation.

Show or say your own response. The adult names back the science they heard or saw. Then choose a next move.

Secure / mixed / misconception / access / method or evidence.

What I said, and what it changed: _____

Next I will: _____

Use your book. Point, say, sign, draw, type, write or use an exact scribe. You may pass. Keep your first answer.

Made from the web lesson SCI-GROW-A2-W05-P2
Version _____

Made by Matt · © Matt Roper · CC BY-NC-SA 4.0: free for classroom use, share and adapt with credit, no resale